

Introduction of computer system

Interface to Operating System

Graphical User Interface (GUI)

- Most common interface for user.
- using mouse and keyboard as input.
- using icon.



Interface to Operating System

Command-Line Interface

- Most common interface for user before GUI be invented.
- Only using keyboard as input.

```
mar@mar$ cat metadata.xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE pkgmetadata SYSTEM "http://www.gentoo.org/dtd/metadata.dtd">
<pkgmetadata>
  <herd>base-system</herd>
  <use>
    <flag name="bashlogger">Log ALL commands typed into bash; should ONLY be
      used in restricted environments such as honeypots</flag>
    <flag name="net">Enable /dev/tcp/host/port redirection</flag>
    <flag name="plugins">Add support for loading builtins at runtime via
      "enable" </flag>
  </use>
</pkgmetadata>
mar@mar$ sudo /etc/init.d/bluetooth status
Password:
* status: started
mar@mar$ ping -q -c 1 en.wikipedia.org
PING rr.esams.wikimedia.org (91.198.174.2) 56(84) bytes of data.

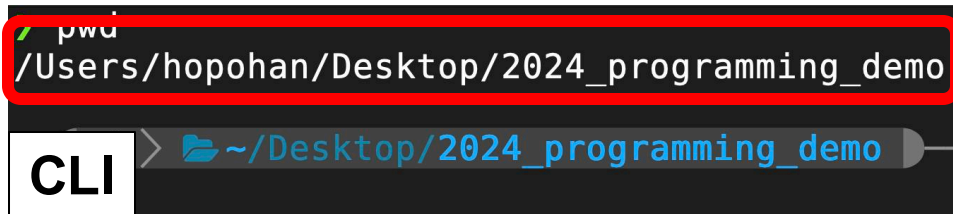
--- rr.esams.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 2ms
rtt min/avg/max/mdev = 49.820/49.820/49.820/0.000 ms
mar@mar$ grep -i /dev/sda /etc/fstab | cut --fields=3
/dev/sda1      /boot
/dev/sda2      none
/dev/sda3      /
mar@mar$ date
Sat Aug 8 02:42:24 MSD 2009
```

Interface to Operating System

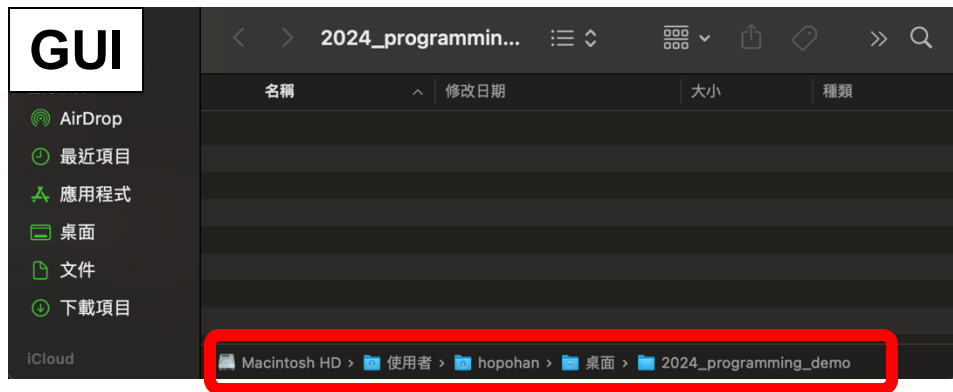
Command-Line Interface Demo

- Check the Path
- Some programs need to be added to the PATH in order to be accessible through command line interface. If you want to check your current path, you can execute the following command:

- PWD



A terminal window with a dark background. The command `pwd` has been entered, and the output is `/Users/hopohan/Desktop/2024_programming_demo`. The output line is highlighted with a red rectangular box. To the left of the terminal, there is a white box with the text **CLI** and a right-pointing arrow.



Interface to Operating System

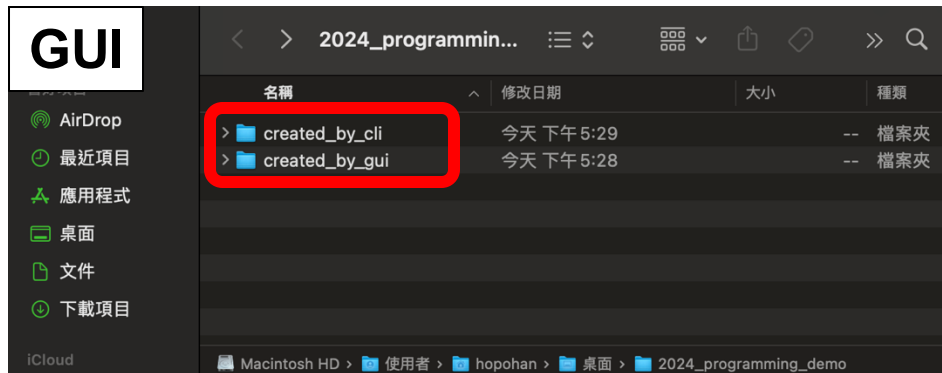
Command-Line Interface Demo

- List files and folders
- If you want to list files and folders in that directory use:

- ls

```
> ls  
created_by_cli created_by_gui
```

CLI



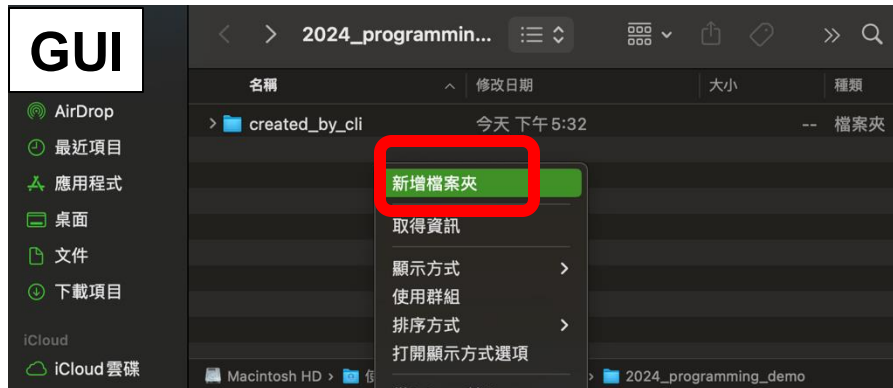
Interface to Operating System

Command-Line Interface Demo

- Create a directory
- Creating a new directory is done using:
 - `mkdir <foldername>`

```
> ls
> mkdir created_by_cli
> ls
created_by_cli
```

CLI



Interface to Operating System

Command-Line Interface Demo

- Remove a file or directory
- To remove a file execute:

- `rm <filename>`

Warning : There is no way to recover files deleted by rm.

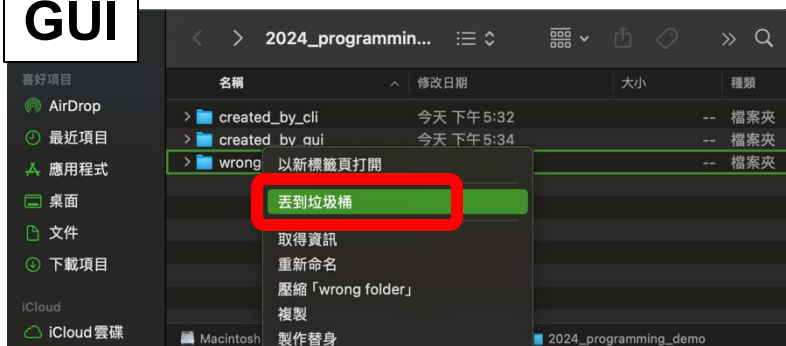
- Removing a directory first requires the directory to be emptied from contents, and then be removed. Removal commands are:

- `rm -r <MyFolderName>`

```
> ls
created_by_cli created_by_gui wrong folder
> rm -r wrong\ folder
> ls
created_by_cli created_by_gui
```

CLI

GUI



Interface to Operating System

System call

- the Interface for program to access.
- Ex:
 - open files
 - read input
 - output message
 - access internet..., etc



Number	Name	Description
1	exit	terminate process execution
2	fork	fork a child process
3	read	read data from a file or socket
4	write	write data to a file or socket
5	open	open a file or socket
6	close	close a file or socket
37	kill	send a kill signal
90	old_mmap	map memory
91	munmap	unmap memory
301	socket	create a socket
303	connect	connect a socket

Open your CLI

macOS

- Command + Space to open spotlight search, type 'terminal' and hit return

Windows (cmd)

- Windows key  + R, type 'cmd' and hit enter.
- type 'cmd' in search bar

Windows (powershell)

- Windows key  + R, type 'cmd' and hit enter.

Linux

- Ctrl + Alt + T

Basic Command (Linux, macOS, Windows powerhell)

command	Description	example
pwd	show current directory	
ls	list directory	ls <dir> or ls -lrt
cd	change directory	cd .. or cd <dir>
mkdir	make directory	mkdir <dir>
mv	move	mv <src> <dst>
cp	copy	cp <src> <dst>
rm	remove	rm <file> or rm -r <dir>
cat	view file content	cat <file>
clear	clear console	

Basic Command (Windows cmd)

command	Description	example
cd	show current directory	
dir	list directory	dir <dir>
cd	change directory	cd .. or cd <dir>
md	make directory	mkdir <dir>
ren	move	mv <src> <dst>
copy	copy	copy <src> <dst>
del	remove	del <file> or del <dir>
type	view file content	type <file>
cls	clear console	

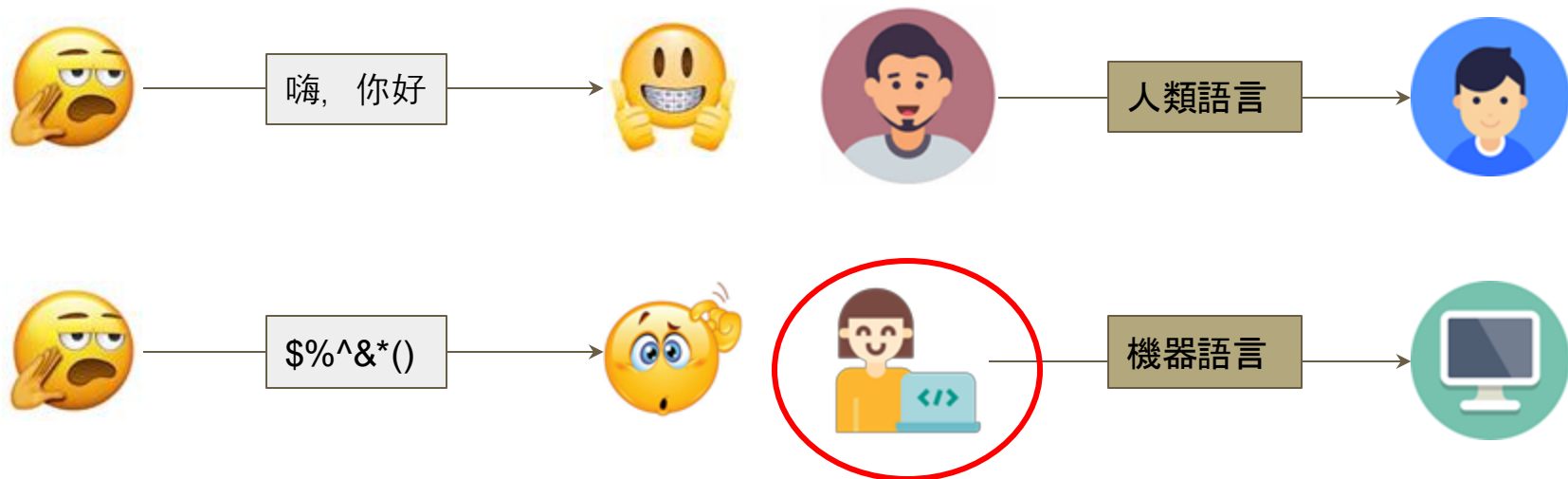
Take away

- What core components are inside your computer
- How each component works
- How your program get executed
- What is Operating System
- Interface to Operating System
- Familiar with CLI basic command

Introduction of Programming language

What is programming language

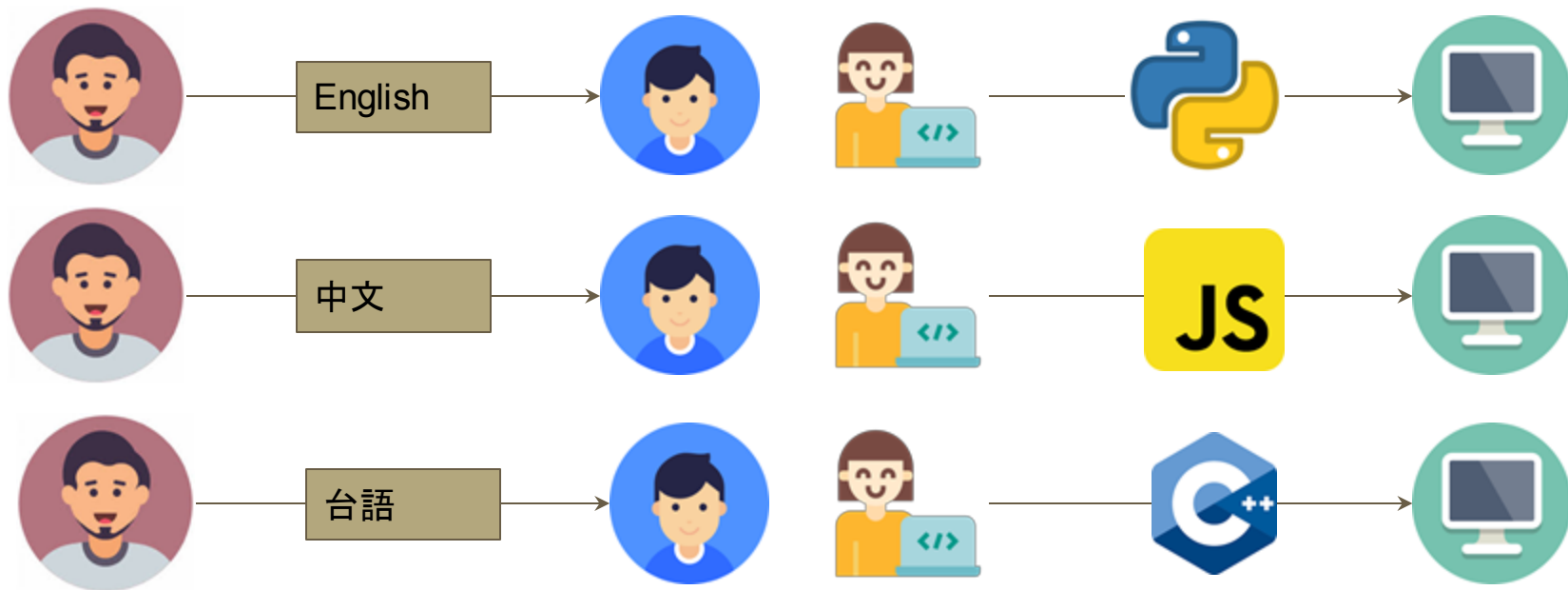
In essence, programming language is a kind of language. Language is a tool that we use to communicate with each other. (見人說人話，見鬼說鬼話)



You're learning a new language !!

What is programming language

Different languages have their own advantages, disadvantages, and purpose.



What is programming language



Python is a **high-level** language for developing high-end applications, such as **AI model**, website, IoT, to name but a few



Javascript is a **high-level** language for developing **frontend website**, all the visual effects on your website are done with it.



C/C++ is somewhere between **low-level and high-level** language. It is primarily used for developing system program, such as **filesystem**, operating system, **device driver**, process/memory management and so on.

Objectness in programming languages



Invented at 1972

Pure imperative language, there is no object concept in the language syntax.



Invented at 1982

C++ is the extension of C. "C with classes" is the core spirit of C++.

But not everything in C++ is object.



Invented at 1994

Everything in python is object. Most of the high-level languages nowadays treat all data, functions as objects.



Invented at 1995

Memory management is handled by JVM. You can design class to create objects.

Objectness in programming languages

We're going to learn python



Invented at 1972

Pure imperative language, there is no object concept in the language syntax.



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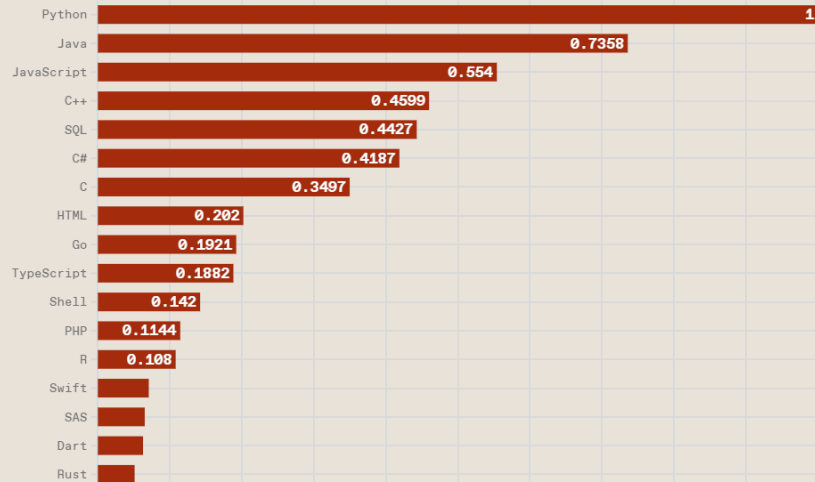
Why learn Python as your first program language?

The Top Programming Languages 2023

Top Programming Languages 2023

Click a button to see a differently weighted ranking

Spectrum Jobs **Trending**



PYPL Index

Worldwide, Aug 2023 :

Rank	Change	Language	Share	1-year trend
1		Python	28.04 %	+0.3 %
2		Java	15.78 %	-1.3 %
3		JavaScript	9.27 %	-0.2 %
4		C#	6.77 %	-0.2 %
5		C/C++	6.59 %	+0.4 %
6		PHP	5.01 %	-0.4 %
7		R	4.35 %	+0.0 %
8		TypeScript	3.09 %	+0.3 %
9	↑↑	Swift	2.54 %	+0.5 %
10		Objective-C	2.15 %	+0.1 %
11	↑↑	Rust	2.14 %	+0.5 %
12	↓↓↓	Go	1.93 %	-0.2 %
13	↓	Kotlin	1.77 %	-0.0 %
14		Matlab	1.63 %	+0.1 %
15	↑↑↑↑	Ada	1.08 %	+0.3 %
16	↓	Ruby	1.06 %	-0.1 %
17		Powershell	0.97 %	+0.1 %

Why learn Python as your first program language?

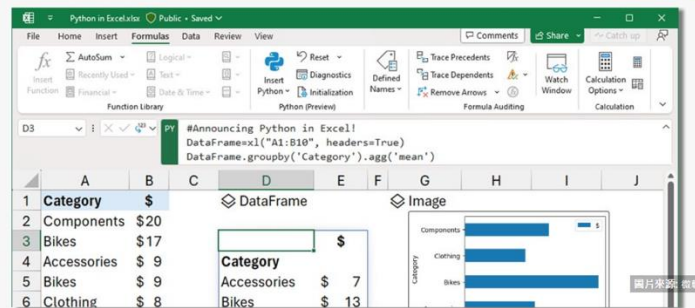
- Microsoft adds Python syntax to Excel, allowing Excel to handle more tasks.

微軟將在Excel加入Python支援，可用於預測分析、機器學習

Microsoft 365用戶將能直接在Excel資料格內，輸入Python腳本程式語言，進行資料視覺化、清洗資料、預測式分析、機器學習等任務

文/ 林妍潼 | 2023-08-23 發表

👍 2,211 分享



專題報導



拓展零信任應用，臺灣駭探購商機



生成式IR來了



CIO看ChatGPT的挑戰和



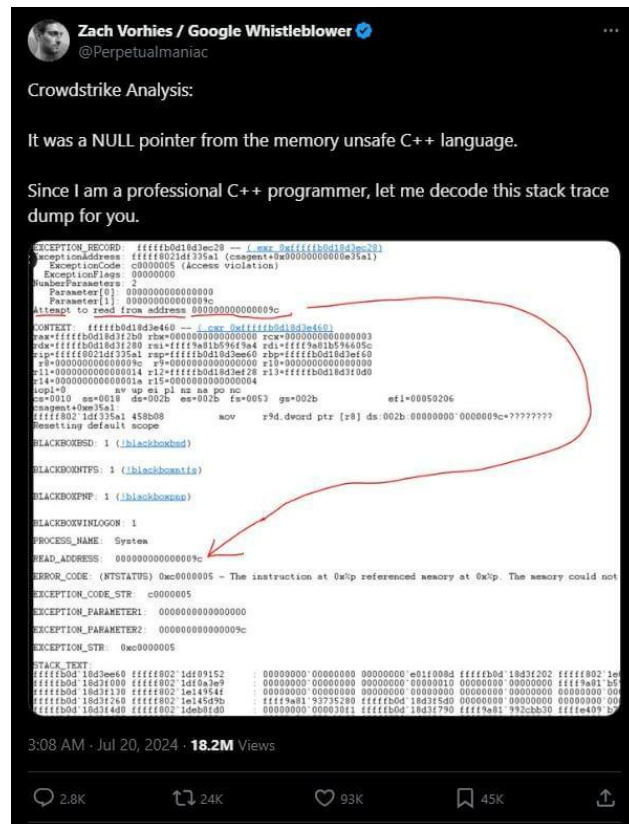
Excel是許多商務人士決策和整理資料的重要工具，微軟本周宣布，將以公開預覽版方式為Excel加入Python支援，讓同一個Excel工作簿能同時使用Python程式及Excel分析公式，允許開發人員利用Excel執行更多元任務，像是機器學習、資料清洗或預測式分析。

Python in Excel目前已經由Beta頻道，釋出給Microsoft 365 Insiders方案成員。

透過Excel加入Python支援，用戶可以在Excel資料格內，輸入Python腳本程式語言，Python在Excel不需設定將執行在Microsoft Cloud上，Python計算會在雲端

Why learn Python as your first program language?

- Although the C language has advantages in execution efficiency, it also brings many issues, such as the global crash that occurred on July 19, 2024.



Why learn Python as your first program language?

- US President Joe Biden's administration wants software developers to use memory-safe programming languages and ditch vulnerable ones like C and C++.

White House urges developers to dump C and C++

News

Feb 27, 2024 · 4 mins

Application Security C Language Programming Languages

Biden administration calls for developers to embrace memory-safe programming languages and move away from those that cause buffer overflows and other memory access vulnerabilities.



CREDIT: MAGDALENA PETROVA

Related content

NEWS

TypeScript 5.6 enters release

With the RC, Microsoft has reversed a change to how the searched for tsconfig.json file

By Paul Krill

Aug 29, 2024 · 3 mins

JavaScript TypeScript Prog

HOW-TO

Static classes and inner classes

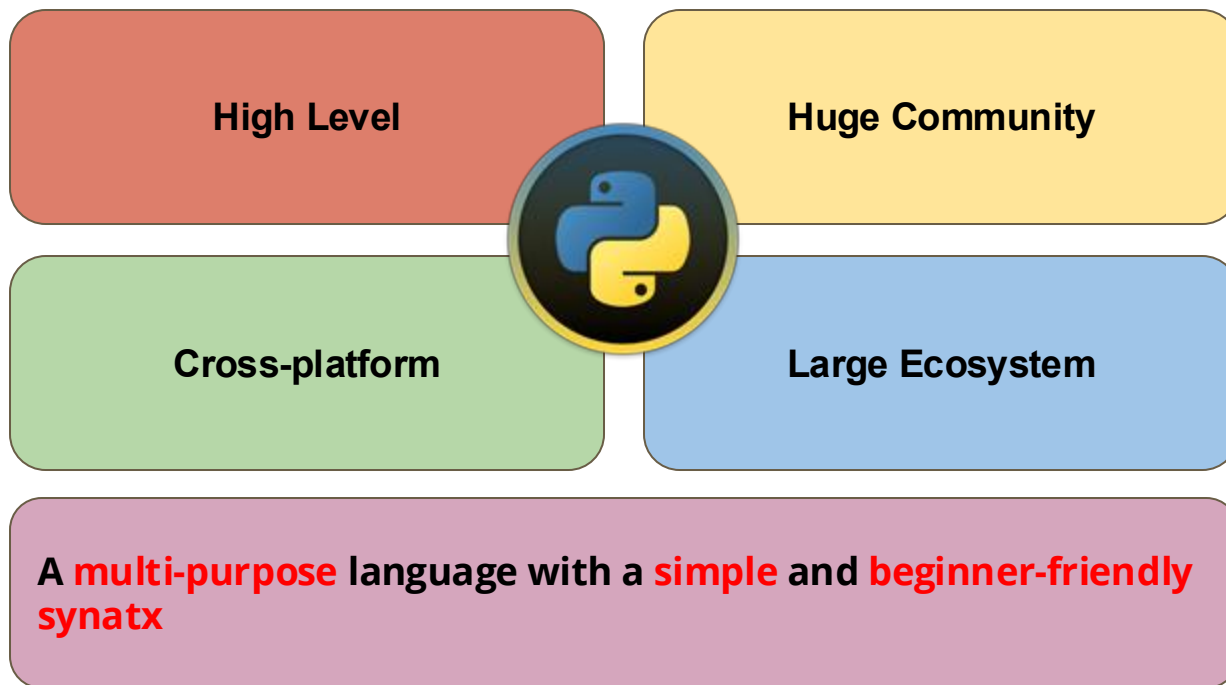
Learn how to use the four types to organize your Java code.

By Jeff Friesen

Aug 29, 2024 · 19 mins

Java Programming Languages

Why learn Python as your first program language?



Python is AI language



Most Popular introductory languages

Rank	University	Python	Java	MATLAB	C	C++
1	Carnegie Mellon	15-110, 15-112				
1	MIT	6.00, 6.01				
1	Stanford		CS106A			
1	UC Berkeley	CS61A				
5	UIUC	CS101	CS125	CS101		
6	Cornell	CS1110		CS1112		
6	U.Washington	CSE140	CSE142			
8	Princeton		COS126			
9	Georgia Tech	CS1301, CS1315		CS1371		
9	UT Austin	CS303E	CS312			
11	Caltech	CS1				
11	U. Wisconsin Madison	CS301	CS200			

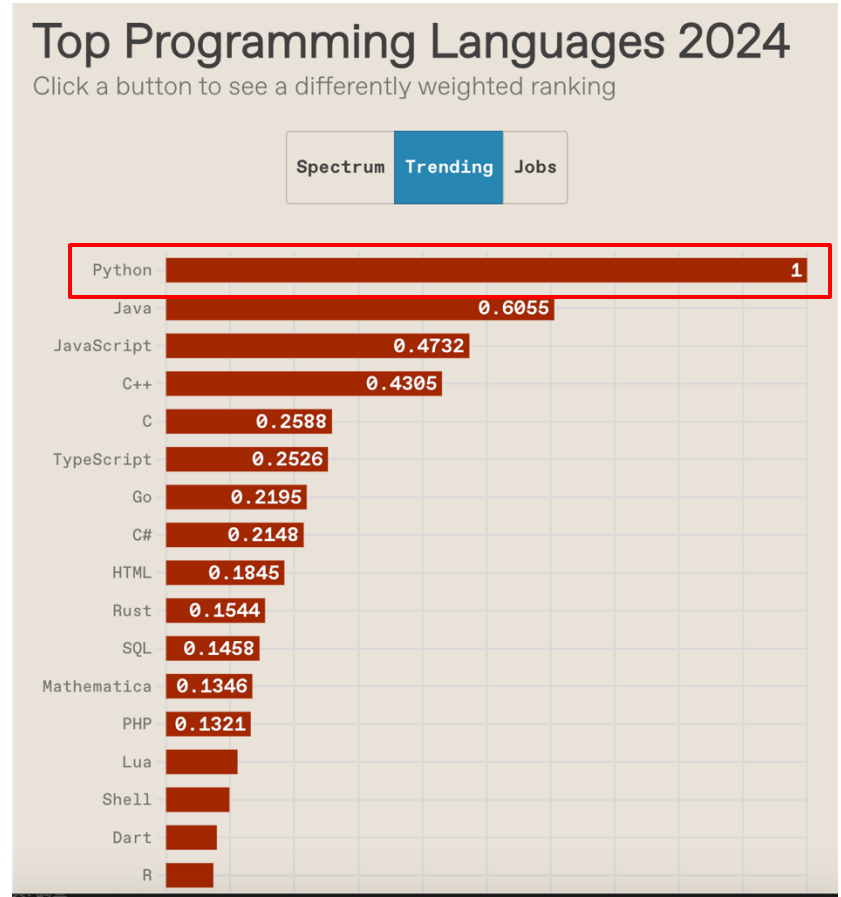


Python is the most popular **introductory language** at U.S top university

Ref: <https://cacm.acm.org/blogs/blog-cacm/176450>

Most Popular languages

<https://spectrum.ieee.org/top-programming-languages/#toggle-gdpr>



Take away

- Programming language is used to communicate with computer
- No specific programming language can fit all needs

END